AAKASH SHARMA

Robotics & Embedded Systems Engineer | Third-Year B.Tech in Robotics & AI, JSPM University Pune

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EDUCATION

JSPM University Pune Pune, India | 08/2024 - Present
Bachelor of Technology in Robotics & Artificial Intelligence (Third Year, Pursuing)
- Core Focus: Autonomous Mobile Robotics, Closed-Loop Motor Drives, Real-Time AI, Kinematics, Embedded C/C++

K.M. Agrawal Jr. College, Kalyan, MH - Higher Secondary Certificate (HSC) in Science & Mathematics 07/2021 - 03/2023

St. Mary's High School - Secondary School Certificate (SSC) | Completed foundational curriculum

TECHNICAL ARSENAL

Microcontrollers & Hardware: ATmega328P, STM32 (ARM Cortex), Arduino Platform, ESP32, Raspberry Pi
Motor Drives & Power: L298N Dual H-Bridge, Star-Grounded Dual-Rail Power, Geared DC Drives, Steppers, Servos
Firmware & Control Loops: Hardware Timer1 PWM, Lookahead PID, Exponential Moving Average (EMA), Non-blocking FSMs, UART, I2C, SPI
Sensors & Instrumentation: LM35 Thermal, SW-420 Shock, MQ-series Gas, Ultrasonic ToF, PIR, Reflectance IR Arrays, Oscilloscopes, DMM
Software, Simulation & CAD: Embedded C/C++, Python, KiCad EDA, SolidWorks, Logic Analyzers, Git

KEYSTONE ENGINEERING BUILDS & PROJECTS

MotorGuard - Smart Industrial Motor Protection & Condition-Based Monitoring (CBM)
[Stack: ATmega328P, Timer1 Hardware PWM, LM35 EMA, SW-420 Shock, 3S Dual-Rail Li-Ion]
- Engineered non-blocking FSM with dual-stage thermal derating: 65% PWM limp-home at 50 deg C, shutdown at 60 deg C (3 deg C hysteresis).
- Built sub-15ms casing fluid ingress cutoff and SW-420 bearing vibration lockout to prevent catastrophic field failures.
- Developed asymmetric (+3 / -6 step) duty-cycle slew limiting in Timer1 ISR to eliminate inrush current spikes.

6WD All-Terrain Autonomous Mobile Robot Platform
[Stack: Skid-Steer Kinematics, 6-Motor Independent Drivetrain, 360-deg Obstacle Array]
- Designed high-torque 6-wheel-drive mobile base capable of navigating rubble and unstructured outdoor terrain.
- Implemented differential left/right bank velocity mixing enabling instantaneous 0-degree pivot skid turns.
- Integrated 360-degree ultrasonic telemetry array for continuous spatial mapping and autonomous evasion.

High-Speed Autonomous Line-Following AGV (Automated Guided Vehicle)
[Stack: High-Density Reflectance IR Array, Lookahead PID Loop, Microsecond ADC Polling]
- Designed front-mounted 5-channel reflectance sensor array with weighted centroid calculation (+/-0.5 mm deviation).
- Tuned closed-loop Kp/Ki/Kd differential steering loop executing at sub-millisecond polling rates for acute track junctions.

Vehicular Alcohol Detection & Active Braking Interlock System
[Stack: MQ Gas Spectrometry, Ignition Cutoff Relay, Progressive ECU Limp-Home FSM]
- Developed automotive safety interlock sampling driver breath alcohol concentration (BAC) in real-time.
- Constructed progressive 3-stage FSM: audible cabin alert -> 50% PWM throttle derating -> full electronic ignition lockout.

Automated Contactless Door Access & Motion System
[Stack: Ultrasonic Time-of-Flight, Dual PIR Arrays, Anti-Pinch Current Feedback]
- Engineered contactless entrance system fusing ultrasonic ToF ranging and PIR motion to eliminate false hallway triggers.
- Implemented armature current-sensing safety interlock tripping instant auto-reversal upon < 1.2A pinch resistance.

CAMPUS EVENTS & PRACTICAL EXPERIENCE

Campus Technical Events - JSPM University Pune Pune, India | 2026
Role: Lead Technical Organizer
- Lead Organizer: Track Titans - Intercollege Line Following Robotic Competition (23rd March 2026).
- Lead Organizer: ROBOTzAI x Monster Energy BGMI Mobile Battle Game (Official Partner: Monster Energy, 16-17 Sep 2026).

JSPM University Pune Pune, India | 05/2025 - 07/2025
Role: Student Intern - Machining & Precision Fabrication
- Completed hands-on training on CNC (Computer Numerical Control) and VMC (Vertical Machining Centre) milling machinery.